

ActInf Textbook Group ~ Cohort 6 ~ Session 19 (Chapter 8, Part 1) 8/26/2024

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 19 (Chapter 8, Part 1) 8/26/2024 | 52:53

okay we're back in cohort 6 for chapter
8 first meeting so enter however you wna
do

it

nice sound was good um yes so welcome

today we're talking about chapter eight

in the 2022 active inference textbook by

friston par and kulo and uh chapter 8 is

following chapter 7 where we discuss

discrete State space models um as we

know the second half of the textbook is

largely centered around actually

designing your own active inference

models as opposed to uh primarily verbal

descriptions of the theory in the first

half so in chapter eight which

complements chapter seven we turn to

continuous State space models which the

authors state are well suited for

modeling physical uations and pinging on

sensory receptors and for the continuous

motion of the effectors EG muscles we

used to change the world around us uh

we're presented with model examples for

the dynamical systems involved in motor

control as well as the concept of

generalized synchrony and finally

constructing Hybrid models for combining

discrete and continuous variables at

differing

levels uh section 8.2 moves on to

movement control uh we're introduces

some notation for the stochastic

equations involved in continuous time

models which we'll notice are a little

bit different from those in chapter 7 um
and then we also differentiate the the
symbols between those of the generative
model and those of the those of the
generative process which is also a
reminder that oftentimes we produce
models whose um construction is actually
rather similar equation wise to those
the generative process an example for
motor control and reflex arcs uh viewed
as descending proprioceptive signals is
given which describes how State
predictions can be pulled towards these
signals like point detractors we also
see how action is involved in uh true
State changes in uh the generative
process to bring outcomes closer to what
the model expects and further how more
complex forms of these equations are
used in more typical use cases uh as
with discrete time models continuous
time active models are used to draw
inferences about the causes of
Sensations B basing and
brain and then uh some simple
nomenclatures so in continuous uh State
space models usually it's X is the new
standin to refer to States Y is now for
observations or incoming sensory data
and U there there are different ways of
of doing it but we also have like for
example you can use an Ω to to
represent um another aspect like
different kinds of fluctuations it
actually starts so one of the equations
starts to look like a like a linear uh
equation model so like a $y = g \cdot \Omega$ plus
 Ω um so kind of like a $y = MX$ plus

b or or an error term for
Omega and there there's quite a bit in
this chapter so I'll just give a bit of
a summation of the rest 8.3 were
introduced to different kinds of
dynamical system which better rep kind
of illustrate the idea of how attractors
come to play in continuous models or or
or shown the uh lka volterra Dynamics
which follow from uh Predator prey
Dynamics researched in say ecology or
zoology with an example of three
different populations plants herbivores
and
carnivores uh showing a kind of
oscillatory pattern between their
populations the idea being that
carnivores eat the herbivores uh who eat
the plants and so they're all kind of
dependent upon one another and they
start to show these kind of kind of uh
non equilibrium like steady state kind
of dynamics that that arise over time
and it's as if there's a these kind of
attracting points that they they arrive
around but never actually land and
settle there um we also have luren
systems talking about things like
criticality um then we'll we'll skip
ahead a little bit Section 8.4 a very
significant Concept in active inference
is generalized synchrony they use a bird
song example for modeling uh to to
illustrate generalized synchrony so the
bird song example for modeling
communication and multi-agent inference
problems uh they're uh the song birds uh
take turns singing so it's a kind of

turn taking problem and it's as if they're singing from the same hymn sheet two birds with similar models learn each other model parameters over time they start to synchronize with one another over time so their internal states and state predictions come to resemble one another reflecting a primitive kind of theory of mind the dynamics are modeled by a higher level whose state predictions of all according to a slow luring system impact the parameters of a faster luring system at the lower level again with action here the birds are singing to minimize prediction errors arising when no song is heard as the luring system evolve modeling chaos the bird synchronization brings about what the authors call an identical synchronization of chaos um this also involves sensory attenuation another very significant concept the birds alternate between two uh between attention to sensory data and sensory attenuation to carry out actions this back and forth with two agents becomes a kind of conversation between them with the generally linear dynamic in their predictions right rising in a lower dimensional space so generalized synchrony can be applied to a range of diverse phenomena from synchronization between internal and external states such as the agent and its environment to communication social behavior and multi-agent settings like like the song bird example where where agents internal

states start to uh sort of mirror those of each other then section 8.5 kind of uh concludes with discussing hybrid models which very well could have deserved its own chapter um but the you know the idea is you could have models that say are hierarchical where one layer perhaps a lower level uh as in figure 8.6 as Daniel was brought up um the lower layer could for example take in continuous data and try and model that and send up whatever kinds of predictions it makes in a different kind of form to a higher level which then makes some kind of you know discreet State space inference about that continuous data and make some kind of inference about it which then is set back down to the lower level um to affect its its its decision making and actions so um I think that's a good enough summary to stop there um and now I will open the floor any questions or points of conversation whenever it was introducing uh having a smooth space instead of a discreet space like the last chapter said it kind of introduced it just like oh yeah this is what you use whenever the other one's impractical this is a little bit empirical but like have you guys found that you want to use like discreet more than anything as it kind of implies in this chapter or do you guys find that like you just pick the one that seems to be right for the data and then you run with it is there a preference or is there a norm that we

use good question um I mean for
from if it's in the form of like a
like a personal question like personal
experience uh I I've largely just work
with discrete State spaces uh I'm coming
from kind of a particular like social
sciences agent based modeling sort of
background where things are presumed to
occur in discret time steps where you
know ideally your data effectively is
just like you know discret logs in a
spreadsheet or something like that um
and so like most of the time I sort of
default to thinking about uh discrete
State spaces that said though um I mean
I've worked with neuroimaging data I'd
love to make some kind of synthetic like
hybrid model that actually pulls in uh
Neuro Imaging data and I would
absolutely switch to a continuous State
space space setting for for that right
basically just I would probably focus on
what kind of data it is that that your
agent is receiving in the format and
then take that approach as opposed to
getting too creative from the start um
yeah yeah another practical aspect is
that pmdp only has discrete time setting
so to use that at this point kind of
bound there um and also separate out the
state spaces from the time handling so
here in the in this chapter they kind of
bring both the continuous time handling
and the continuous dates space like they
go really well together but you also
could have like a continuous variable
like just between zero and one any
number in the discrete

time however then it's like if you think about that it's like the it's like the Matrix math that we looked at in chapter 7 doesn't apply where so if we just had four locations but location could have been a continuous variable um so I think the the fullest slash uh most technologically enabled would be to go with the data type if you're collecting from an analog sensor then like EEG or a continuous reading or the continuous location of something however even continuous measurements at some point on digital Hardware are getting sliced up and so then it's like okay how much do you want a coar grain um that influences a lot of the handling it's just simpler with discrete State space and discrete time okay cool thank you that kind of reminds me back in college we would uh talk about scraping electrons off of ccds and actually extracting that information and how it all is digital at that level gives me nightmares uh I also wanted to ask could you talk a little bit more about the uh Lorent part after the uh volara thing I didn't quite understand that as well as I understood the uh volara explanation and that is the rather I almost feel like just despite how Central the bird song example is used in this chapter I actually think it's it's the Loren system that that shows some much more complexity

um the kind of complexity involved in in
some of these kinds of models but
um I I'll I'll um yeah go for go for it
no if you if you had something sorry I'm
just like yeah so okay so the the Loren
is kind of a classic chaotic system it's
kind of the um grandparents chaotic
system and that was Loren studying the
weather and saying wait even closely
related trajectories diverge that's the
leap exponent all that how it's being
used in the chapter is to give a bu
stability to turn taking so it's like
here if you if all you knew was the x
value you'd find okay I'm staying
between I'm staying negative and then
it's like whoop now there was a phase
change into being positive so that like
sort of by stable nature of the luren
attractor system is being used to be a a
a
chaotic um switch on turn taking in
Birds um it's it's also kind of relevant
like the birds are not sharing new
information with each other so it's not
exactly like turn taking in say a normal
person conversation it's more like
they're just wanting and preferring that
The Song Remains the Same that the
song's played out and then it's like as
soon as it's being listened to and then
when the
listening um is like wait there's a
missing note then the other bird steps
in um that
that's
yeah go for okay no I'm sorry I was just
going on to say that that's what they now

it makes sense that they were talking
about the synchronization of chaos that
really clicks for me now

yes this is oh the since like 2015 when
I think the fith and friston bird song
comes out it's been used a lot um this
32.3

or 32 series one of the rare ones with a
DOT three um this paper goes into more
detail let's just look at a few of the
key pieces

here

so here here's um sampling from a loren
attractor so you have you can have a
continuous smooth differentiable chaotic
system meaning sensitivity to to small
inputs highly Optive exponents and then
you could sample probabilistically from
that so you're getting sampling that's
like kind of speckling approximating
that um smooth curve um what is done
with the smooth curve so is separate out
on this landscape of Attraction where's
the gradience from higher to lower
density and then what is the solenoidal
flow that goes on an ISO

Contour so then um this is another key
aspect of the continuous time in
discrete time to look any time steps
into the future you need to do these
explicit modelbased rollouts like
actually considering future time steps
and all the combinatoric explosions that
accompany that and could you huh could
you please remind me what an ISO Contour
is okay ISO Contour is like when you're
hiking a line around the hill at the
same elevation okay so it's just

orthogonal to the the the gradient of
steepest elevation is like putting a
ruler on the that the steepest possible
you're on a place on the hill steepest
possible gradient and then you have
orthogonal to that you have around the
the map okay the Taylor series expansion
is when you can take

you the first order approximation is
just the
value second is the slope the red line
third so as you take these higher and
higher approximations you can
approximate arbitrary functions going
out

um here they couple two chaotic systems
they have them um and then they like in
that state space that's described by the
movement of them they show that you can
apply that hel hols

decomposition and say like where what
are the what are the circulatory
flows and then what are the gradient
flows um this this is the um this is
what the system looks like so this is
the two birds so to speak here's their
interface and so system a and system B
and then these kinds of plots are are
verifying like yeah some States like the
third and the sixth their correlation
drops to zero whereas other states
remain

correlated

[Music]

um so this the whole reason why they
have Loren is just show you can put any
smooth curve in the continuous it
doesn't have to be just like a

sinusoid um or anything like that you can have any kind of smooth differentiable curve and sample probabilistically from it so you could generate Point data from an underlying curve and you could take sampled Point data and then regress essentially against and get to a smooth differentiable setting and which one of those directions is useful just depends on what kind of data input you have okay cool that reminds me a lot of very classical machine learning whenever you consider your data points to be you know just points on a manifold and you're trying to learn the shape of that manifold that's okay that's kind of familiar to me yes and so then then sampling based approaches are trying to make a smooth differentiable landscape from the Speckles because with a smooth landscape you can do like gradient descent but you couldn't do gradient descent on just a cloud of points yes absolutely well not well yeah can jump from point to point um and so that's what is getting played with I mean on a bigger scale in the second half of the book but then explicitly in the last part of chapter 8 when they come together in the Hybrid models is like you and and then as Andrew highlighted like temperature in the world may be a continuous variable but you could model it with integers or you could model it with um .1 like one

significant figure of of decimal
Precision so then those are all modeling
choices um and that's the kind of
infinite branching Paths of how you
model
something
okay cool thank
you it's it's also in in the
communication style models so kind of
bird song um EP epistemic communities
like the kind of Twitter hashtag version
on Andrew social science
simulation what becomes really
interesting is like what rails are put
in and then what is actually the
semantics of what is
communicated I'm not gonna lie that's oh
I'm sorry yeah go Ah that's exactly what
I was thinking about whenever you said
like how you divide up temperature like
specifically how you divide it up seems
like it would directly go into how you
are going to interpret the results so
for the bird songs and stuff like that
like just jumping into what frequencies
you're naturally attuned to focusing on
what axioms are naturally put into the
system they are going to represent what
you are going to more easily represent
that was a poorly worded sentence but
it's going to lead you to learn certain
things easier which seems like it would
help with the
uh synchronization of chaos specifically
yeah what will be like forever beyond
that model's Horizon like if you're only
doing temperature by the
tens that but perceiving it by the tens

like one significant figure then there won't be it it won't be able to capture at the ones place whereas if you had a model that was at the decimal point maybe it could kind of do something at a at or like what what kind of communication amongst agents is being set up like if you just say if you have put everything in place so it's like zero means go left and one means go right well then they're just going to be sending ones and zeros and using that to coordinate their movements but then how would you actually make a a kind of packet that convey move left but without putting the rabbit in the Hat so it's unfair so that they're just kind of hinting each other what they already know versus are you trying to explain the sort of um puristic signaling amongst agents that already have a shared semantics or on the other extreme end are you trying to model the emergence of shared communication semantics from some kind of a simpler substrate or like the birds like they already have the him sheet so they're keeping up the duty of the song being played and then you can show really interesting phenomena like in in the papers where they'll show okay bird one will transfer its parameters to two then you remove one you add three and then two can transfer to three so you can say there's a propagation of the fine tuning in this kind of genetic context where it

knows the song but then that model
couldn't be used to address how does the
song change within or between
Generations this is a
little we might have discussed it the
last time we discussed this this chapter
in a previous cohort um so like
page
164 um still withs example there's a
figure
8.5 that uh represent like the
synchronization and communication
process and on the left we have the the
generative model and the attractor
system yeah then on the right
synchronization manifold so so the the
upper graph is supposed to represent
like there's it's a little bit more a I
don't want to use the word chaotic but
you know it's a little less coordinated
um but then after learning there is
organization where it's as if you know on
the Y AIS that's the second
bird um xaxis is the bird so there's
this kind of secreation such that you
know it looks like
AAR you know positive correlation but
what's interesting to Simply that the
values um the value range on the X and Y
axis um are much larger in the second
graph and you know I would need to know
much more about parameter settings Etc
but I I was trying to figure out like if
the birds are like further expanding
their exploration of the state space as
they continue to learn right because
they if I don't know why that would
expand outward um and I I might just be

missing something very simply but it's just it's kind of interesting that it's such a smaller State space that they're awkwardly uh exploring before learning and then after learning it's broader yeah also in both cases it goes further to the right like here it never you know it never goes below 50 I mean never goes below -20 but then it it way overshoots um there might be something like a conservation of like in IA so it's like if you had no synchronization you'd have a circle I mean the path would be traced within a circle and then um if you have total synchronization it's almost like that's kind of like making this ellipse that might project out further but what are these really about well they're second level expectations why is the sensory data that's the actual note that's played into the environment level one I believe is the vocal cord of the bird that's like the warbling of the vocal chord control I don't know if it's a two note song I don't know if there's any auditory representation of these songs that would be pretty fun too level two then would probably represent like who's supposed to be singing and maybe the zero point is the switch over so then then this represents a an extreme belief like that it's plus 60 like 60 to

one odds or e to the 60th or something
that the second bird is
singing but yeah these these Big Blocks
kind of get thrown out and it's like
what are they
do that for
one kind of the
Classic this gets to the predictive
coating
question so I guess the first layer
is the neural control of
warbling and then the second layer
is higher vocal
Center and then it's kind of interesting
like you know is the audio um is there
like a mirror neuron thing where you're
responding to your own generation of
speech I mean there has to be because
you kind of confirm that you're speaking
as you expect but then this was going to
be my question for you Andrew was
why do you think there's sensory
attenuation during
action oh why not just keep High
attention on the sensory inputs while
you're
acting sure my my sense of that from
what I've understood because that's one
of the most seemingly
counterintuitive Concepts uh that I
found in in active inference at least
kind of like in a you know the way it's
presented in the textbook but um the way
I kind of imagine the logic of it in a
more colloquial sense it's like if
I if I am absolutely certain that my arm
is exactly in the position that it's in
and I maintain that sort of belief uh

perpetually it's going to be very hard to change my mind that it is or could be anywhere else right there's a kind of like like by sort of by attenuating that incoming sensory evidence that it's exactly where it is right now um that allows me to kind of like put on pause so to speak the the feeding of that sensory information into reconfirming a belief I have that is there such that now I can actually move it what I what I need to happen is to believe that it's actually where I want it to be rather than where it currently is such that I can actually realize that sort of expectation um I don't know if any of that tracks but that's kind of you know in a simple sense like kind of how I've been understanding since re attenuation yeah I mean I remember we had many fun conversations about getting out of the chair and about disorders of initiating and terminating movement so it's kind of like discrepancy is resolved by changing your mind or changing the world so updating belief States or updating action States if we had complete total rigid confidence that and this also relates to a a a like a key multiple role for the expectations about sensory input they're both trying to be veridical like representing sensory inputs otherwise it's just it's just off kilter and and deceptive so on one hand the expectations have to be um at least tracking real world changes like we're experiencing the

visual generated expectations hence the color and the sharpness and the periphery all that so from a first past like sensemaking perspective you want that visual expectation to be accurate but then if you had total Precision on what that were you wouldn't have the openness of the flexibility to take in New Or different information but that's just on the sense making now we get into the action so then if you had because there's also this role for expectations in pragmatic value because they're also the preferences about sensory inputs so if we're getting if we're getting confirmatory propri receptive input that the arms in position one then there's there's there's no discrepancy to even resolve we we expect it where it is we prefer it where it is so the best the most veritical and the highest pragmatic policy is to do nothing so then there has to be this like transient alleviation of the Precision to even enable the possibility that that um counterfactual about where the arm could be and the propriate reception that a companies that like for that to even be plausible the Precision has to be widened to include it so making things significantly less precise on on reading your incoming sensory data um that said not completely

gone I almost imagine like a sort of
intermittent like a almost like a
blinking like is like U you know as
opposed to completely continuous intake
of of the propri receptive signal of
where your arm is it's like kind of
almost have to like Loosely check and
make sure it's going in the direction
you want it to not to imply that any of
this is necessarily like a conscious
process but but it's as if you need to
kind of intermittently check that it's
going the way you want to and and is in
the position you want it to be in um
else like I could imagine that getting a
little Haywire if the entire process
were paused to the entire duration of
the movement but I might be off base
here with how you're describing it but I
see what I I do see this is the ni lead
end of talking about
decision yeah I mean it's so it's so
well integrated with fixed eye gaze
which is what the example that comes up
again again at the end of the chapter
just briefly fixed eye gaze High
precision and then it's like you're
driving on a windy road it's like pull
off the Precision make the move but you
don't see a blurry set of you don't see
the blur it's like you're just not
paying attention while the Cade is
happening and then as soon as it's fixed
again then the precisions re-engage and
it looks clear and then shorts of memory
kind of like just irons out the gap
with your expectations I suppose yeah
yeah

yeah it's such a I don't know it's such
a almost like a fun concept to like rra
your head around um but it but it does
make I mean it makes sense to me as far
as the the other aspects of like the
theoretical aspects of the active
inference go it's like if you just yeah
if you're fully
fixed on a particular belief that you
have about say the position your arm or
where your eyes are Etc the ideas I mean
you would be self-evidence you would not
apt to
change like the location that you're
looking at uh right so it it's coherent
despite the kind of seeming like you
know what is that how does it work when
first presented with it um and it does
track
to the way that I understand it and
please let me know if you guys like
agree or disagree because I might be
misunderstanding is whenever I read that
I kind of got the impression that like
inside of us there are multiple systems
or at least this is the way I imagine it
there's multiple systems there's like
the proceptive system there's like my
eyes telling my head where my arm is and
there's a whole bunch of things so every
time I take an action there's going to
be all of systems which are engaged and
these systems want to have a lot of
harmony with each other so there's going
to be constant checks every time there's
going to be an action we want to use
this attenuation to make sure okay are
all the gears shifting as they should is

this transmission moving in Step so all of these systems are going to align themselves with every action so there's going to be some model in my head of the world that I see that is going to be built up by the things that I touch and feel especially through Pro reception through sensory stuff uh through up my eyes like my retinas actually take in we I think we went two chapters ago about how the eyes get like such small amounts of information and we actually build up the world around that and a few other things which are going to be a system of things and then the connection of the systems are going to build up this world model that I have in my head and that world model that I have in my head is the thing that we're using the free energy principle with that's the thing that we're minimizing a surprise with so does that kind of read that like you want to constantly have this attenuation to have these checks these systems are in stability in harmony yeah I mean the free energy minimization process applies within a modality and in the composition of modalities so you're right that for a real organism there's also these cross calibration checks um that that might do a lot of the work where in a single modality it doesn't need to be it doesn't need to be as resilient within a single modality because you can like double check you could look at something and

then you could like move it to Double Or
you could see what it appears to be and
then touch it to get confirmatory
evidence from a different and then if
something is showing up coherently
across
modalities it's EXT it's like multiple
confirmatory lines of
evidence absolutely
yeah and this actually leads into
another question I had which is not so
related to chapter 8 but it's in chapter
eight so I'm going to bring it up it's
the question of what do you guys think
about like having multiple ideas on the
same thing kind of the superposition of
ideas and just kind of picking out just
wisping out the one that makes the most
sense in this chapter it talks about the
accurate belief that I am not moving in
order to uh predict the sensory
consequences of movement uh blah blah
blah but I find that really interesting
because it just says the accurate belief
that I am not moving and that could mean
a few things that depends on the frame
of reference which I'm putting it that
depends on the context for the actual
place that I want to apply that free
energy because like I'm not using it
right now but I'm standing on a
treadmill so I could be moving but at
the same time relative to my desk I'm
not moving but then again relative to
the sun I'm absolutely moving but then
whenever you think about all these other
things there's going to be this kind of
like which place do I want to put my

sensory input in my sensory input is not
just my sensory input it's the sensory
input with the context of the
environment I'm building it around where
the environment I'm building it around
right now is this funny little idea of I
am not moving with respect to what do
you guys have any thoughts on that is is
that as deep as it goes is that just
like I read a sentence too hard or
[Laughter]

this is it precisely the kind of uh you
know the the kind of conundrum that that
I was trying to express when I first
started speaking about sensory
attenuation you know it's uh it's so I
keep saying seemingly counterintuitive
because at this point there's something
intuitive my brain that's telling me I
think I get it now um but really I mean
it's tough with um I mean to to be on a
treadmill the difficulty with that that
example is that like if you're just
standing on you know a treadmill let's
maybe go with an escalator uh like if
you're just standing there like you're
not moving like you're moving and that's
accurate in a in a kind of you know
external you're moving in the world's
you know XYZ coordinate 3D World sense
but you're not taking action you know so
if if your mark on blanket is is you
you're not taking the actions to move
your body and at least in the way that
we're saying that and so it's I mean
it's not going to be I don't know if
this necessarily plays into that however
if you're you know standing in place and

you want to move right then like the idea of sensory attenuation is that you would have to sort of downtune or widen the Precision uh or or or otherwise uh regarding your sensory observations of standing in place at least to some you know degree oritt or however you know how it's your choice how you want to actually model that and try to get at that but however it is the ideaas you have to kind of put on pause this idea that you're just standing in place lest you just kind of reconfirm the belief that you're just standing in place you'll even if you like want to move uh that that's not going to sit well with like the rest of your like so-called beliefs as as an agent right the rest of your beliefs telling you well both of my legs are neither of them are moving and like in terms of all my senses I appear to not be moving um self- evidencing like you're G to stay there um you know even if you consciously have some belief that you want to move I mean I I think it'd be I I think it's interesting to look at that from a kind of you know psychiatric standpoint as well or looking at you know kind of phantom pain or or things like that um but uh so so you kind of I mean I think I think the way that Daniel put it earlier too with the sad I think that was really good actually like you're not actually noticing the blur necessarily whenever you move your eyes right it's kind of like like you know we predictive coding sort of think of it is like the higher

the higher order is just kind of like
descending and put like expectations
like kind of fills in the The Gap
whenever you move your eyes around but
you're not actually focusing on that FL
you're really downtuning kind of that
sensory information you'd otherwise be
taking in because you're not focused on
oh my eyes are exactly in you know
position

2.21 or 2.1 by 3.1 now they're in 2.0 by
3.0 like you're not necessarily traing
every incremental bit that you move
right you're just I'm looking here I
want to look here um everything in
between is just like a kind of skip and
your your mind Fs in the blank so it's
not too terribly jarring your everyday
experience right also surprise
minimization the treadmill separates
which you know not necessarily an
evolutionarily um prevalent stimulus
separates out two senses of I am there's
I am making movement which could be
proprioceptive change so rate of
proceptive change is I am making
movement then there's I am moving
distance

which would relate like you said a lot
to the reference frame and then those
two might have it's like if you were in
a dream and you felt like you were
running but you felt like you were
moving

slow that could be like a gear
differential between the appropriate
receptive that you felt like you were
like working hard then the visual that

you weren't and then that would be
interpreted as like a Salient
differential yes
absolutely and also I I think dreams are
just deeply fascinating in the realm of
uh active inference that's just amazing
to
me the dreams will be super fun to
explore and like on the lucid dreaming
like I like can they
control and reduce their agency level
like can they submerge or can they can
they control their attention or could
they you know what aspects of the
generative
model are being are being
modified and what are the bounds of the
Lucid
agency what are they reducing or
increasing their Precision on and then
what then are there different like
subtypes where where somebody might be
able to remember more but they can't
have agency over wanting to remember
more or something like that I don't know
oh that's that's a really really cool
idea how much of the because you are the
environment in that case there's no the
separation between you know the self and
the environment you're in a dream it's
just it's all you so that is such a
fascinating angle to go
down yeah but at the same time the way
you're processing that it's as
if you're you're a mar on blanket inside
of a Marco blanket as supp
you know you're you're still distinct
from you uh the world that your mind's

creating I almost wonder if there's a if
a dream is just some kind of like
residue of of offline structure learning
right like there's just there's some
kind of mashing around of different you
know model parameters or however you
want to say and then suddenly you have
this visual scene where you get to like
move around in their affordances and at
some point you might just notice that
you have more control than what you
think you do I don't know fun
conversation had of run around dreams
always the hybrid model at the end that
was kind of most um most recently
unpacked in the uh active inference does
not contradict folk
psychology a nice careful title not that
it that it supports or or verif IES any
kind of but just that it that it isn't
the wrong link I'll I'll fix that later
but just that it's that it um doesn't
contradict and then the belief desire
intention folks psychological model
um and so just
like
can if all things can be modeled in
active inference then that would include
these other kinds of experience
situations and other
things so then like then again that kind
of reflects well what what is it that
allows us to describe anything are there
things outside of our modeling
scope or are there just situations that
are just empirically very challenging to
assess like you couldn't get a verbal
self-report from somebody in a lucid

dream or just something like that like
there's situations where you couldn't
get a data point but you could model
where you could get a data
point I'm personally of the mindset that
our tools can lead us to places way
outside of how creative we can be
because whenever we're like building
these things we're you know we have a
very serious like we have our math hats
on and I think of physics whenever they
found like black holes and stuff like
that and the big bang and whenever they
looked at it they said that's completely
ridiculous there's no way that's true
and I think that with a lot of tools we
see very similar things to where we
didn't well it led to very unintuitive
results yeah yeah U want to you
know I think it new to apply active
inference in certain areas that I'd like
to you know apply it to but uh like a
followup point on that is that you
know so like a tend to like for
centuries models that really simplify
the idea of what is human behavior or
something like utility or profit
maximizers you know and and the kind of
outcomes model that using computational
usually what ends being very identical
to one another it's kind of a zero some
game Cut Pro to cut and because people
are so identical they don't have their
own preferences They Don't Really trade
things with one another it's just a very
flat simple environment and uh suddenly
once you once you include things like
like uh you know allow for more

complexity suddenly you you end up
having agents who trade with one another
in ways that leads to Market volatility
and and crashes and Bubbles and all of
these other kinds of things so what's
interesting to me is like on the one
hand I do agree with you I think that
these kinds of tools lead to all kinds
of like crazy new discoveries that I
mean that's one of the greatest as the
combination of like epistemic and
pragmatic value there is like doing
research to make new discoveries and at
the same time it's like sometimes I
think these tools will actually help us
understand already exist existing
phenomena but now we actually know how
to recreate them and now better
understand what their initial causes are
too so um just coming at it from kind of
the other angle as well no I completely
agree feeling like you get to see
something from a new angle using a new
tool is actually one of the most
satisfying things I think you can
experience and now cognitive science is
moving that it's not just we're taking a
picture of like our elbow that we
couldn't have seen it's like oh that's
what my elbow looks like from that angle
if it's like that's what my mind looks
like from that angle or that's a view of
a mind that hasn't
existed then then that is just so
strange loopy and
trippy
absolutely yeah I'm trying to remember
there's some famous quote from uh I

don't know if it was Richard finan or
who but uh some quote like if I can't
create it and I don't understand it it
was it's like a kind of
wor by a model if I cannot create it I
don't understand
it so trying to understand things
chapter
seven that's where we're at we I wonder
where I got that
from we understand creating the discrete
time models I think it's fair to say at
this time we understand creating the
discreet time models more not not like a
some fundamental aspect of them just
that we've had a lot more experience
with pmdp and the discreet time maths
are a little clearer but then the
continuous time especially with RX and
fur it'll be very cool to kind of flip
in and out and make both and Nest them
and and just really start to see
like how how and where they can be can
we could just do a continuous in a
discret time or state space for like
every single situation and just kind of
have like two models in parallel then
just recombine them as
needed yeah yeah absolutely I I I
imagine like uh you know I I was
thinking about um how to model like for
example agents who do have access to
some kind of you know uh re some kind of
continuously measured resource and like
what's important is to be actually
actually be able to measure if something
is like higher or or lower like actually
having that kind of like order ordinal

scale or otherwise because that isn't that kind of the issue with trying to take some continuous value and then bending it such that becomes discrete choices is that you also have to somehow include the logic that the range of 0 to 10 is necessarily less than the range of 11 to 20 right like if you if you have a model that doesn't quite get that then it's going to make associations like it might make very odd looking associations between what it means to be in 0 to 10 versus 11 to 20 right so um anyway maybe conversation for another well next time we Explore More eight maybe we'll look at and possibly add some questions eight getting you know eighth inning after the seventh inning stretch um some some have left the stadium um but but then we we head back into the data oriented nine and the closing 10 so cool thank you fellas see you next time wonderful talk to you guys thanks soon peace bye take care